

A4: High Fidelity Prototype and Documentation

CCT305H5 Design and Implementation of Multimedia Documents

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Group 10

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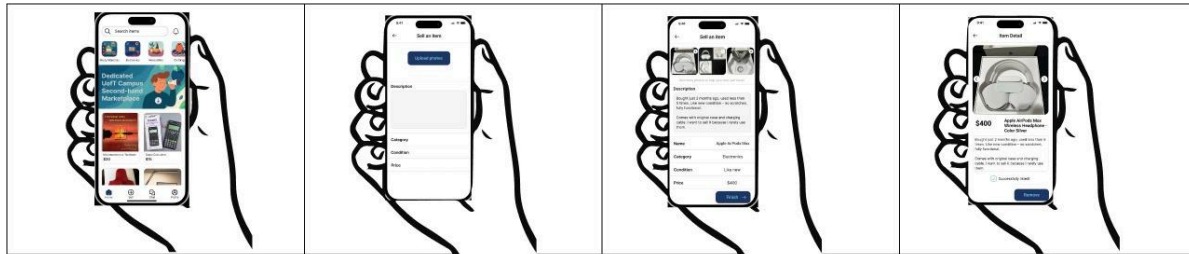
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Apr 1, 2025

- **Link to clickable High-fidelity prototype - Cedric Zhu**

<https://www.figma.com/design/9Qr1Rv8DgE5tvILgSrbzjU/CCT305-A4-Hi-fi-Prototype?node-id=0-1&p=f&t=jJE8DECaIqiBr2HF-0>

- **Story Board - Jiayu Song**

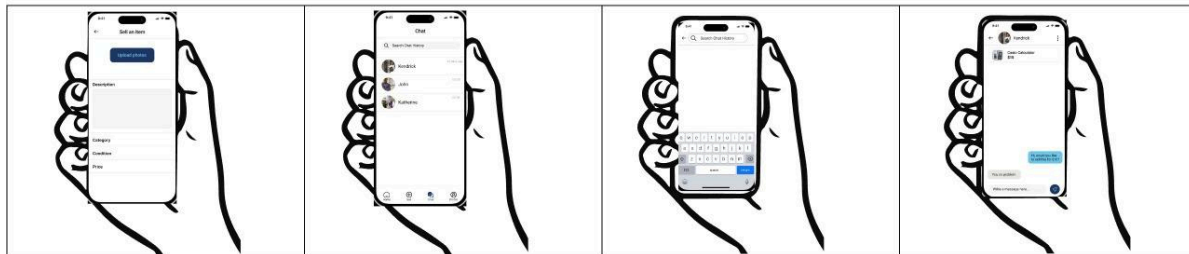


Home page
User opens the app and lands on the homepage with featured items and navigation bar.

Sell
User clicks the "Sell" icon in the navigation bar to start creating a new listing.

Upload
User uploads photos and begins entering item details such as title, category, condition, and price.

finish
After filling out all the fields, the user clicks "Finish" to publish the listing.

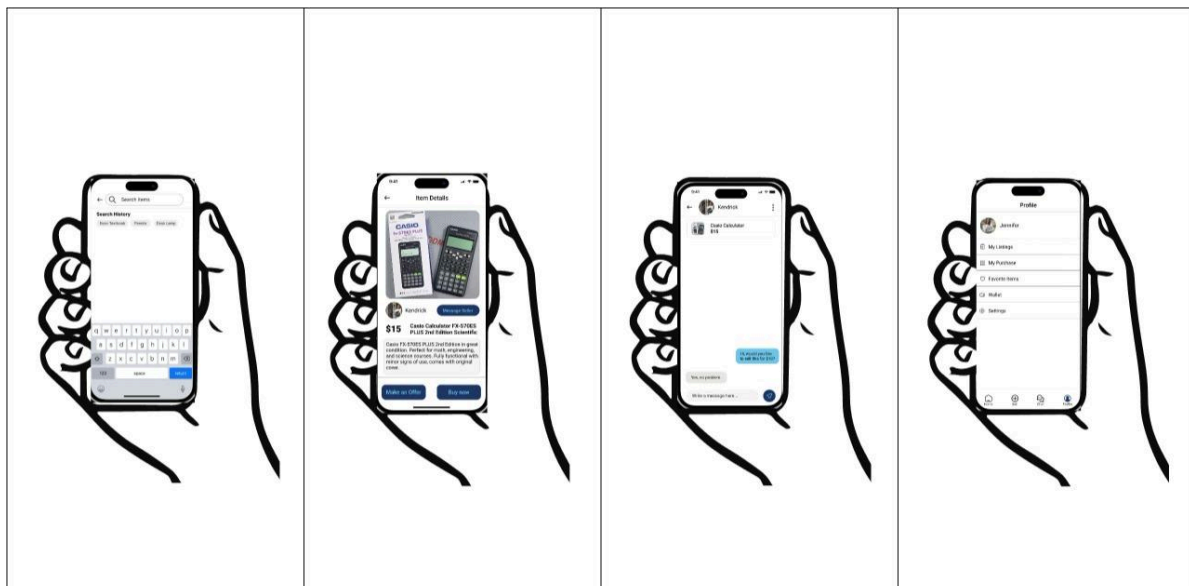


Remove
User can now view the posted item and has the option to remove it.

Chat
User clicks on the "Chat" tab to view recent conversations with sellers.

Search chat history
User uses the search bar to find previous conversations by seller name or item.

Chat
User opens a specific chat to view conversation history and send new messages.



Search items
User types keywords into the search bar to find specific items.

View Item Details Users can click on a product listing to view detailed information, including the price, seller, condition, and description. "Make an Offer" or "Buy Now"...

Messaging seller
User opens the chat window and starts a conversation with the seller to negotiate price or arrange meet-up.

Profile
User visits their profile page to access personal listings, purchase history, favorites, wallet, and settings.

- The Usability Task - Antong Du

The usability test at this last stage focuses on the three essential user tasks evaluated in A3: (1) listing an item as a seller, (2) searching for an item as a buyer, and (3) messaging a seller as a buyer. These are all core components of the user journey on our second-hand trading platform and remain unaltered to provide consistency in the evaluation. Each task has been updated in the high-fidelity prototype in response to feedback to enhance usability.

1. Listing an item as a seller

The old version had users create a listing by uploading images, entering product details, choosing the category, condition and the price. Some users said that the option of upload was too plain, and how the order of the fields was not very clear.

In the high-fidelity prototype, we rearranged the listing page for better clarity and ease of use. Each input section (photo, title, description, category, price, condition) is now clearly separated into cards with added visual icons with helpful supporting text. We also added a progress indicator at the top to show users which stage they are in in the listing process. Along with this, validation warnings and character limits help prevent input errors. These additions, though small, make the process feel more structured and user-friendly.

2. Searching for an item as a buyer

This task made the user search for an item using the search bar and filters. In A3 it stated that people had issues with inconsistent page transitions and filters being reset and wanted more fluid interaction.

For the high-fidelity version, we implemented a persistent filter bar and a swiping animation to browse through different pages. Filter choices such as the price range, category, and the

designated meet-up location will no longer be hidden in another page but will remain visible and active. In addition to this, not only are results displayed in consistent card format, with seller verification included, along with the price. These are all changed to increase the fluidity of one's browsing experience and the visual organization of these tasks. This task now feels more intuitive than before without changing its main steps.

3. Messaging a seller as a buyer

The main reason users came to the chat system was to begin a conversation with a seller. "We like how you integrated messaging into the platform but dislike how you put search high up on the right side of the screen and that there are no unread messages."

Then the high-fidelity prototype moves the chat window back and groups them together based on whose activity is most recent. We moved the search bar for the chat somewhere underneath the section of the user navigation below it. So instead of having to look a long way left before you can find chat history. Something that could be fixed right away. Also, if unread messages come in, a red dot notification is now added at the bottom of the nav bar, along with suggested quick replies. This helps get a clearer line of communication and makes it an easier task to assist. These changes simplify the messaging task while maintaining its core structure.

- Changes Made and Results of Usability Testing – Dejian Zhang

During the high-fidelity prototype stage, we improved several core functions of the platform based on the feedback collected during usability testing. These changes directly addressed the pain points users experienced when completing key tasks and significantly improved the overall user experience.

First, for the process of sellers posting goods, we redesigned the information entry interface. The original upload page in the prototype was fully functional, but the disordered field order and dense interface information caused users to get lost during the entry process easily. Based on the test feedback, we divided the page into several clear modules: image upload, title input, product description, category selection, price setting and item condition. Each part is presented in the form of a card, with icons and prompt texts to make the process more guided. These changes not only improve ease of use but also make the process more structured and more fulfilling.

Second, chat system optimization solves the problem of unreasonable interface logic. The original design placed the search bar at the top of the chat page, and users reported finding chat records inconvenient. We moved it to the bottom of the navigation and sorted the conversations according to their activity, improving information access efficiency. These enhancements give users a greater sense of control when communicating and enhance the professionalism of the platform.

Third, we updated the overall font and color of the prototype. Since we are designing a second-hand trading platform on campus, we used the blue color of the University of Toronto as the background for the buttons that interact with users (HEX #1E3765). The font in the buttons is (HEX #6FC7EA). The product text and application background are displayed in white (HEX#FFFFFF) and black font (HEX#000000), clearly showing the goods and content.

In short, all changes were closely based on the honest feedback from user testing to make core functions smoother, more intuitive, and safer. This reflects the importance we place on user experience and lays a solid foundation for the eventual implementation of the platform.

- User Feedback We Rejected – Dejian Zhang

We carefully analyzed every suggestion throughout the usability testing and user feedback collection process, but some feedback was deliberately rejected or postponed after team discussion. One of the rejected suggestions was the introduction of an anonymous chat function. Some users wanted to hide their identities during the initial communication stage to increase privacy. However, since our platform is positioned as a system based on real-name registration within the campus to establish a trusting relationship, anonymous chat would weaken this foundation of trust. It may lead to irresponsible behaviours such as no-shows and malicious price-cutting. In addition, the survey results show that most students prefer to use the real-name method authenticated by their school email to enhance the sense of security in transactions. Therefore, we have chosen not to adopt this suggestion to maintain the transparency and protection of transactions.

Some users also suggested restricting offline trading locations to only allow trading in designated campus areas to improve transaction security. Although we agree with the rationality of this suggestion, the team ultimately decided not to impose restrictions but instead to use “recommended trading locations” as an alternative. This decision was mainly based on user flexibility: while restricting the transaction location can help with security management, it may also increase the user's burden and inconvenience, especially when the item is large or geographically dispersed. Therefore, we chose to display the recommended transaction location on the product details page instead of imposing a mandatory restriction on the location.

In summary, most of the suggestions we rejected were due to development feasibility, the balance of user experience, and the prioritization of functions. These suggestions are not invalid, but they are unsuitable for implementation at the current prototype stage. Based on the platform's development pace, they may still be integrated and optimized in the future. After laying a solid foundation for the product and verifying the stability of core processes, these expansion suggestions can be an essential reference for later optimization and expansion.

- Hypothetical Next Steps - Antong Du

If we were to carry on with another round of design our main priorities would be to implement an in-app payment system that includes escrow functionality that each party could trust. While the high-fidelity prototype allows users to simulate item exchanges, it still falls short of equally protecting the buyer and seller against fraudulent transactions. For the next development we would implement a way for each party to hold a payment temporarily until the buyer is able to confirm that they have received the item in good condition. This form of protection was greatly requested during testing and is essential to building long-term user trust. We would also add a visible transaction progress tracker for both parties to monitor the status of the deal in real-time.

Another path we would take is the addition of an exhaustive rating and review system to maintain community trust. The existing version shows user verification and basic transaction history, but development for the future would include written reviews and star ratings after each transaction. With this addition, reviews would only be shown after both users are done

submitting their input into the system, or if a certain amount of time goes by for a review to appear. As users are given the option to submit their ratings to buy and sell goods, they will gradually accumulate a reputation profile in the form of trust badges and other visuals to display all feedback given out through the interface. Having a fully developed profile, buyers and sellers in the future would feel more at ease when deciding whether a specific item is worth purchasing.

And finally, we would begin development for a fully native mobile application to support push notifications, camera access, and even being able to save drafts offline. Despite our prototype imitating enough mobile behavior through touches and shapes, only an actual app can achieve the kind of responsiveness and real-time interactivity that is expected by those using it. Through a mobile application we could add features such as displaying meet-up locations based on a user's location, instant notifications for unread messages, and faster listing creation through native phone tools. Our target audience is of course students who wouldn't go anywhere without a mobile device, so having a native app developed would exponentially improve daily use and incentivize the continuous use of our platform.

- What your group learned about design, user research, usability - Jiayu

Song

Throughout the phases of designing, testing, and refining this high-fidelity prototype, our team developed a greater insight into design principles, user research, and usability. We incorporated course concepts into our design; initially, we recognized that effective information architecture plays a crucial role in the overall design, enabling users to accomplish tasks more effectively. We applied the card block model to

streamline the product release process, resulting in a more organized and logical layout for users.

Moreover, we realized how vital colour selection is in the overall interface design. We selected the blue associated with the University of Toronto as the primary button colour, creating a strong contrast against the white background, which improves both readability and interface recognition while reinforcing the identity of our campus platform to enhance users' trust in it. Furthermore, the addition of interactive features like icons and prompt texts enhances usability and guarantees that users receive timely, clear feedback.

Regarding user research, we discovered that user feedback guides us in making design choices centred around their needs. Usability testing revealed unforeseen issues and encouraged us to consider the platform's usability from various viewpoints of actual users, highlighting concerns like the poorly positioned search bar. We recognized that users have diverse information requirements while engaging with the platform, prompting discussions on how to meet these needs more effectively. Regarding usability, we found that even minor adjustments can significantly enhance user efficiency. We learned to evaluate the practicality of feature suggestions in relation to the platform's positioning; for instance, we chose not to implement suggestions like “anonymous chat” or “mandatory transaction location” as they would negatively affect the platform’s trust mechanism and flexibility.